2017-09-23

CRF Layer on the Top of BiLSTM - 2

Review

In the previous section, we know that the CRF layer can learn some constraints from the training dataset to ensure the final predicted entity label sequences are valid.

The constrains could be:

- The label of the first word in a sentence should start with “B-“ or “O”, not “I-“
- “B-label1 I-label2 I-label3 I-…”, in this pattern, label1, label2, label3 … should be the same named entity label. For example, “B-Person I-Person” is valid, but “B-Person I-Organization” is invalid.
- “O I-label” is invalid. The first label of one named entity should start with “B-“ not “I-“, in other words, the valid pattern should be “O B-label”
- …

After you read this article, you will know why the CRF layer can learn those constrains.

2. CRF Layer

In the loss function of CRF layer, we have two types of scores. These two scores are the **key concepts** of CRF layer.

2.1 Emission score

The first one is the emission score. These emission scores come from the BiLSTM layer. As shown in figure 2.1, for example, the score of w_0 labeled as B-Person is 1.5.

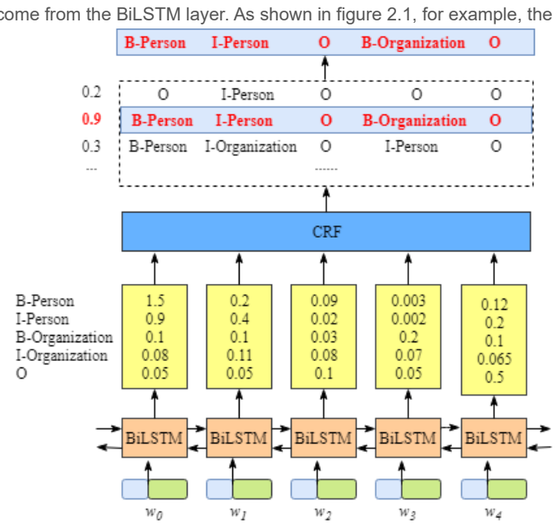


Figure 2.1: the emission scores come from the BiLSTM layer

For convenience, we will give each label a index number as shown in the table below.

Label	Index
B-Person	0
I-Person	1
B-Organization	2
I-Organization	3
O	4

We use x_{iy_j} to represent the emission score. i is the index of word and y_j is the index of label. For instance, according the figure 2.1, $x_{i=1,y_j=2} = x_{w_1,B-Organization} = 0.1$ which means the score of w_1 as B-Organization is 0.1.

2.2 Transition score

We use $t_{y_iy_j}$ to represent a transition score. For example, $t_{B-Person,I-Person} = 0.9$ means the score of label transition $B-Person \rightarrow I-Person$ is 0.9. Therefore, we have a transition score matrix which stores all the scores between all the labels.

In order to make the transition score matrix more robust, we will add two more labels, START and END. START means the start of a sentence, NOT the first word. END means the end of sentence.

Here is an example of the transition matrix score including the extra added START and END labels.

	START	B-Person	I-Person	B-Organization	I-Organization	O	END
START	0	0.8	0.007	0.7	0.0008	0.9	0.08
B-Person	0	0.6	0.9	0.2	0.0006	0.6	0.009
I-Person	-1	0.5	0.53	0.55	0.0003	0.85	0.008
B-Organization	0.9	0.5	0.0003	0.25	0.8	0.77	0.006
I-Organization	-0.9	0.45	0.007	0.7	0.65	0.76	0.2
O	0	0.65	0.0007	0.7	0.0008	0.9	0.08
END	0	0	0	0	0	0	0

As shown in the table above, we can find that the transition matrix has learned some useful constrains.

- The label of the first word in a sentence should start with “B-“ or “O”, not “I-“ **(The transtion scores from “START” to “I-Person or I-Organization” are very low.)**
- “B-label1 I-label2 I-label3 I-…”, in this pattern, label1, label2, label3 … should be the same named entity label. For example, “B-Person I-Person” is valid, but “B-Person I-Organization” is invalid. **(For example, the score from “B-Organization” to “I-Person” is only 0.0003 which is much lower than the others.)**
- “O I-label” is invalid. The first label of one named entity should start with “B-“ not “I-“, in other words, the valid pattern should be “O B-label” **(Again, for instance, the score $t_{O,I-Person}$ is very small.)**
- …

You may want to ask a question about the matrix. **Where or how to get the transition matrix?**

Actually, the matrix is a paramer of a BiLSTM-CRF model. Before you train the model, you could initialize all the transition scores in the matrix randomly. All the random scores will be updated automatically during your training process. In other words, the CRF layer can learn those constraints by itself. We do not need to build the matrix manually. The scores will be more and more reasonable gradually with increasing training iterations.

Next

2.3 CRF loss function

Introduction of CRF loss function which is consist of the real path score and the total score of all the possible paths.

2.4 Real path score

How to calculate the score of the true labels of a sentence.

2.5 The score of all the possible paths

How to calculate the total score of all the possible paths of a sentence with a step-by-step toy example.

References

[1] Lample, G., Ballesteros, M., Subramanian, S., Kawakami, K. and Dyer, C., 2016. Neural architectures for named entity recognition. arXiv preprint arXiv:1603.01360. https://arxiv.org/abs/1603.01360

Note

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zpeng

5 years ago

Could you expalin why add the start and end flag in detail? Thanks.

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Ersin Yar

9 years ago

Thank you for this great tutorial. I have a question related to the joint training of LSTM and CRF layer. You learn transition matrix in training phase. In general, one would obtain predicted labels and calculate a loss function using true and predicted labels. As far as I understand in CRF layer calculation of loss function is done using true path and all other paths. So, in training phase we don't predict an output sequence (using viterbi) and we don't calculate a loss function using only true and predicted path. We calculate the loss function using true path and all other paths. Then, we optimize the parameters of CRF and LSTM using back-prop from this loss function. Is this correct? Is this the way you implement?

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createmomo Mod Ersin Yar

9 years ago

Hi Ersin, many thanks for your comment and sorry for my late reply. Yes, you are right. In the training phase, the paper used the true path and all the other paths. To be more specific, in the loss function, the paper computes two scores: the score of the true path and the total score of all the possbile paths including the true path. The general idea of the loss function is we are trying to improve the real path score ratio. (You could find the details in following articles).

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Ersin Yar createmomo

9 years ago

Thank you for your reply. I also want to ask about transition scores. At a first glance, I thought that transition matrix includes the transition probabilities between different states (per -> loc). But, I guess that is not true. The example you give above does not show probabilities. It gives nonnegative scores. Is there any other constraint on the transition score matrix other than non-negativity?

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createmomo Mod Ersin Yar

9 years ago

Hi Ersin, in fact, there is no constraint on the elements of the transition matrix in the paper listed in the reference section. Some people called the elements in the transition matrix probabilities, but I prefer to call them scores which could be negative or positive. In other words, these scores are learnable parameters as the same as the weights (i.e. W) in a model. We could believe that the BiLSTM-CRF model will learn reasonable values of these scores by training from train data set.

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Y

Yida

9 years ago

Very clear explanation and super helpful, thank you!
BTW, if the transition matrix is initialized randomly, and suppose the training data is NOT enough for the model to learn well, then finally the model may predict impossible label sequences such as B-Person -> I-Organization. What to do if this happens? Thanks!

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createmomo Mod Yida

9 years ago

Hi Yida, it is great that you found the explanation is helpful. As for your questions, my answers may be not absolutely right, but I am happy to share my opinions: 1) If the training dataset is not big enough, you can just try traditional models firstly such as CRF model. You may get better performance than neural network based models. 2) But if you really want to use the deep learning methods, you can compute the transition scores between labels manually based on your training dataset and use these scores to initialize you transition matrix; 3) Probably just fix the transition matrix by your manually computed scores, in other words, to set the transition matrix is not learnable.

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Y

Yida createmomo

9 years ago

I see. I'll try what you suggested and thanks for your help! :-D

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D

dy l

9 years ago

Looking forward next section!

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createmomo Mod dy l

9 years ago

Thank you very much for your support!

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